

NLP Deliverable 2 Guide

1 Introduction

The objective of this project is to apply machine learning techniques to a Named Entity Recognition (NER) task. Algorithms to solve NER tasks are very useful in many contexts, from information retrieval to question answering systems.

When solving our NER task, we will confront two different machine learning approaches: the classic machine learning approach based on handcrafted features and modern machine learning, where the features are automatically extracted from the training data by the learning algorithm.

2 Deliverable zip file

This project has to be submitted by a single member of each group using a specific task created in the webpage of the course. The project needs to be in a **zip** file containing all the code to reproduce the results. The **zip** file containing all the work needs to have the following form:

- **name1.name2.name3.zip**. Where **name1,name2,...** are the names of the members of each group. Please write your full name. Example: If your name is “John Doe” write **JohnDoe**.
- Groups can be up to 4 people. The project documentation should contain a section briefly describing how the work has been distributed between the group members.

3 Basic rules of the deliverable

- The **zip** file needs a **main.pdf** file with a description of your work.
- The **zip** needs a notebook **reproduce_results.ipynb** that loads the work and evaluates it.
 - Accuracy on the train set and test set but taking into account only those predictions where the ground truth is not “O”.
 - Confusion matrix on the train and test set.
 - F-score (see https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html).
 - The predictions for the TINY_TEST (look below) and the accuracy in such dataset.
- The data used for this project is **train_data_ner.csv**, **test_data_ner.csv** containing training data and test data respectively.

- Since the project may be too heavy, it should be sent by email to `daniel.ortiz@ub.edu` by a single member of each group. In the email, you may provide a link to dropbox or similar.
- In addition to this, the document `main.pdf` should be submitted also by a single group member to the corresponding task in the virtual campus.
- The **deadline for the project will be the 20 of june at 11:59 pm.**
- The `zip` file does not need to include the training data (but you can include it if you want).

Each requirement from this document that is not satisfied in your work will imply a reduction on your final mark.

4 Project structure

Your project must be in a folder with the following tree structure:

```
> name1_name2_name3
  > main.pdf
  > data
    > train_data_ner.csv
    > test_data_ner.csv
    > tiny_test.csv
  > train_models.ipynb
  > reproduce_results.ipynb
  > fitted_models
    > model1
    > model2
    ...
  > utils
    > utils.py
  > environment.yml
```

Some details:

- `main.pdf` is a document containing a general description of your work and your results. More details about this can be found in [Section 4.1](#).
- `train_models.ipynb` is a notebook containing all the code required to train the models and store them in `fitted_models`
- `reproduce_results.ipynb` is a notebook that must: load the data, load the fitted models from disk and evaluate the models, following the rules for the deliverable described in this document.
- `fitted_models` is a folder where you will save your trained models. Any model used in `reproduce_results.ipynb` should be imported from `fitted_models`.
- `utils` is a folder containing functions that might be used in the `train_models.ipynb` and/or `reproduce_results.ipynb`. The goal is to keep the notebooks clean (avoid writing large functions inside the notebook, if you do it you can take them out, write them in `utils.py`, import them and use them).

- `environment.yml`: environment file to ensure that your code is runnable in another machine. The same kind of file that was requested for the first deliverable. It can be created by activating the desired environment and then executing

```
conda activate environment
conda env export > environment.yml
```

4.1 Content of `main.pdf` file

The `main.pdf` file should contain a description of the carried out work, including the results obtained. The document content should be defined so that it can be read independently from the rest of the submitted files.

The file should have the following structure:

1. Document front page with title and names of team members.
2. Table of contents.
3. Introduction (problem to be solved, general description of the data and main techniques that were applied, etc.).
4. Exploratory data analysis (description of the data statistics that are considered relevant).
5. Methodology (brief description of the techniques applied to solve the task).
6. Experimental setup (training configuration for the different models, evaluation metrics, etc.).
7. Results.
8. Conclusions.
9. Team member contributions.
10. Use of AI.

4.2 What models to test

You need to test:

- The CRF model explained during the course. You can think about adding or modifying feature templates to improve results. Additionally, you can also try the structured perceptron model. In this case, you can use the code provided in the corresponding course notebook.
- At least one deep learning model.

4.3 TINY_TEST

Notebook `reproduce_results.ipynb` should print the predicted tags for the following TINY_TEST in the format

w1/t1 w2/t2 w3/t3 w4/t4

where `wi` is a word and `ti` the tag associated to `wi`.

```
# TINY_TEST (these sentences and the ground truth labels are given in the
#           tiny_test.csv file)

The programmers from Barcelona might write a sentence without a spell
checker.

The programmers from Barchelona cannot write a sentence without a spell
checker.

Jack London went to Parris.

Jack London went to Paris.

Bill gates and Steve jobs never thought Microsoft would become such a big
company.

Bill Gates and Steve Jobs never thought Microsof would become such a big
company.

The president of U.S.A thought they could win the war.

The president of the United States of America thought they could win the war
.

The king of Saudi Arabia wanted total control.

Robin does not want to go to Saudi Arabia.

Apple is a great company.

I really love apples and oranges.

Alice and Henry went to the Microsoft store to buy a new computer during
their trip to New York.
```